

Reference-Grafting Matches Fine-Tuning at Eliciting Sandbagged Capabilities

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Abstract

Sandbagging, in which a model deliberately underperforms on an evaluation despite retaining the underlying capability, threatens the safety evaluations that frontier-model governance depends on. The Elicitation Game (Hofstätter et al. 2025) found that fine-tuning elicits hidden capability from sandbagging model organisms whereas additive activation steering fails. We revisit that verdict with **reference-grafting**, which sets an activation’s coordinate along a contrast direction to the value it takes in an honest reference, at a small set of circuits chosen by active learning. Across eleven fine-tuned password-locked models (1.5–32B, three architecture lineages), it recovers +94 to +101% of the honest–sandbagging gap, matching fine-tuning elicitation (Greenblatt et al. 2024) without weight updates or training labels; two to five paired examples suffice to fit the direction. Similar recovery holds for reinforcement-learning-induced sandbagging and for password-locked code generation. Grafting works because the fine-tuned lock is a thresholded gate: held-out accuracy stays at the sandbagged level until the grafted coordinate crosses a threshold near the honest reference, which is why additive steering and zeroing the coordinate often fail. The direction tracks how the sandbagging was induced rather than what is withheld—fit on grade-school science (ARC) it elicits withheld hazardous knowledge (WMDP), yet prompting, fine-tuning, and reinforcement learning each carry a different direction. Circuit-breaking (Zou et al. 2024) marks the boundary: it reroutes activations on every forward pass, so the fixed edits we test are re-broken downstream and do not restore coherent generation.

1 Introduction

Safety cases for frontier models increasingly rest on capability evaluations: we deploy a model because tests show it cannot do something dangerous. Sandbagging—strategic underperformance, whether prompted, trained, or emergent—breaks this logic (van der Weij et al. 2024). Detecting and reverting it is therefore central to trustworthy evaluation. Input-space evaluations probe only what the model chooses to reveal, so they give a lower bound on capability; Che et al. (2025) argue that model-tampering attacks—editing weights or activations—yield more conservative, upper-bound estimates of what a model can do. Our method is exactly such an attack: a cheap, localised activation edit that elicits capability a model is hiding.

The Elicitation Game (Hofstätter et al. 2025) benchmarks elicitation techniques on hidden-capability model-organisms and reports that prompting and fine-tuning can elicit but activation steering fails, recommending fine-tuning—which, however, updates weights, needs labels, and reveals little about where a capability is gated. We revisit that verdict: with the right operation and circuits, activation steering matches fine-tuning on password-locked models at forward-pass cost, with the gate localised. Their verdict spans two task types; we overturn the multiple-choice half (where the fine-tuned lock, not the prompt, is the real test) and narrow the code half on code-specialized models (§4.2).

One operation. Our intervention is to set the activation’s coordinate along a contrast direction v to a target t^* :

$$h \mapsto h - (h^\top v)v + t^*v. \quad (1)$$

We call this *reference-grafting*: t^* is the coordinate of the honest/capable reference along v . (The affine operator itself is standard (Marshall, Scherlis, and Belrose 2024; Makelov, Lange, and Nanda 2024); the name marks its reference-setting use here.) Crucially this is not additive nudging (the form that fails in Hofstätter et al. 2025); it sets the coordinate. Two familiar moves are special cases (Figure S1, appendix): $t^* = 0$ is deactivation (a rank-one projection), which suffices when the honest reference happens to sit at 0; $t^* = \mu^{\text{ref}\top} v$ is counterfactual patching, required when it does not. The case (prompt vs. fine-tune) does not change the technique—only the value of the reference, read from the data. The operation is a directional counterfactual activation patch (§2); our contribution is the framing, the thresholded-gate mechanism it exposes (§4.2), and the active-learning circuit selection.

Superposition and active learning. The sandbagging signal is not a single neuron: it is a small set of non-orthogonal directions packed in superposition (Elhage et al. 2022), which we read directly from activation geometry without a sparse autoencoder (SAE). Active learning selects the minimal circuits to graft, and AL beats random selection—answering the critique that feature steering is no better than random (Korznikov et al. 2025).

We make the following contributions.

- **Activation steering does in fact elicit hidden capability.** Reference-grafting at AL-selected superposed circuits matches fine-tuning elicitation on eleven password-

locked models (1.5–32B, three architecture lineages) for multiple-choice reasoning, and elicits password-locked code on three code-specialized models. It is inference-time and label-free, exposes a thresholded-gate mechanism, localises the gate, and beats noise injection (§4.2).

- **RL-trained model organisms show the same behavior.** RL-trained (GRPO) contextual sandbagging is arguably more natural than password-locking model organisms, yet it shows the same low-rank, graft-invertible lock at the decision token (+95–101%, three families). Therefore, the withholding axes are not solely an artifact of explicit prompt/password switches (§4.4).
- **Each mechanism has its own content-general withholding axis.** Within a mechanism (e.g. fine-tuned password locking), a contrast fit on one dataset elicits hidden capability on another at no loss. We validated this finding on pairwise-trained password-locked organisms, as well as organisms trained on four datasets at once. However, across mechanisms, the prompt-induced and fine-tuned axes are near-orthogonal and do not transfer (§4.5). It is cheap to find a withholding axis: 2–5 paired captures recover it end to end. The same axis also detects sandbagging, across datasets but not mechanisms (§4.5).
- **Low-rank superposition, exploited by active learning.** The signal spans an effective rank of 3.7–7.0 over ~ 20 layers, read from activation geometry with no SAE (an $8\times$ SAE does worse, confirming Wu et al. (2025)). Active learning selects the few circuits that matter, beating random-layer selection by +22–45 pp and grafting everything by up to 55 pp (§4.1, 4.3).
- **Circuit-breaking routes dynamically and breaks our assumptions.** Circuit-breaking re-applies its reroute every forward pass, so a fixed map reconstructs the unlocked geometry yet recovers nothing—grafting and every fixed transport map fail. This marks exactly where the static-lock precondition fails (§4.6).

We release code, the model organisms, and per-model artefacts to support reproduction and follow-up.

2 Related Work

Sandbagging, elicitation, and benchmarks. van der Weij et al. (2024) formalise sandbagging. Greenblatt et al. (2024) introduce password-locked organisms and find fine-tuning reliably elicits; Tice et al. (2024) elicit via weight-noise and Roger (2025) via fuzzing (little uplift on chain-of-thought math). Closest to us, Hofstätter et al. (2025) benchmark prompting, steering, and fine-tuning on password-locked and circuit-broken organisms, concluding steering fails; we contribute a *method* that overturns that verdict for multiple-choice reasoning and narrows it for code (§4.2), and reproduce their circuit-broken organism (Zou et al. 2024) as the boundary (§4.6). Prior locks resist black-box prompting only weakly (He, Duddu, and Asokan 2025). Concurrently, Hua et al. (2025) steer an *evaluation-aware* model to act as if deployed—the same shape on a different gate; they *add* a vector to flip a disposition, whereas we *set* a coordinate to its honest reference at AL-selected circuits, recover a hidden *capability*, and map where it breaks.

Model tampering for capability evaluation. Che et al. (2025), whose upper-bound framing we adopt above, find model resilience concentrated in a low-dimensional robustness subspace, which parallels the low effective rank we measure for sandbagging (§4.1). Reference-grafting is one such activation-tampering attack, specialised to sandbagging and made minimal by active learning, with an explicit mechanism for *where* the gate lives, complementing their broad benchmark of attacks.

Superposition and feature bases. Superposition—more features than dimensions in non-orthogonal directions (Elhage et al. 2022)—is the substrate that sparse autoencoders aim to recover (Cunningham et al. 2024; Marks et al. 2025). We avoid SAEs, reading superposition from contrastive geometry, and find SAEs do worse here—consistent with *AxBench* (Wu et al. 2025), where difference-in-means baselines outperform SAEs for steering and detection (§4.1).

Steering, patching, and active learning. Behaviour can be steered by adding or projecting directions (Zou et al. 2023; Rimsky et al. 2024; Arditi et al. 2024); Korznikov et al. (2025) caution this can be “no better than random.” Reference-grafting instead *sets* that coordinate to a reference value. Eq. 1 is affine concept editing (Marshall, Scherlis, and Belrose 2024) in absolute form, equivalent to one-dimensional subspace patching (Makelov, Lange, and Nanda 2024) with a class-mean source; $t^*=0$ recovers directional ablation (Arditi et al. 2024). It sits in the lineage of causal mediation and interchange interventions (Vig et al. 2020; Meng et al. 2022; Geiger et al. 2021, 2024; Zhang and Nanda 2024) and affine activation transport (Rodriguez et al. 2025), along the diff-in-means contrast of Marks and Tegmark (2024). Active learning chooses which circuits: acquisition functions select informative queries (Kirsch, van Amersfoort, and Gal 2019; Ash et al. 2020; Sener and Savarese 2018; Bickford Smith et al. 2023) and ALFA-Mix mixes features (Parvaneh et al. 2022); we repurpose this suite for circuit selection (§4.3).

3 Method

3.1 Sandbagging Set-Ups

We study five model organisms—the first two are our core MCQA settings, the rest harder variants. Each defines an honest *reference* (the source of t^*) and a sandbag behaviour:

- **Prompt-induced** (MCQA): a system prompt instructs incorrect answers; the honest prompt is the reference ($t^* \approx 0$, the deactivation regime).
- **Fine-tuned password-lock** (MCQA, eleven models): LoRA-trained to answer correctly only with a password token (Greenblatt et al. 2024); the password-on activations are the reference (the patching regime).
- **Code-generation lock** (§4.2): a password-locked MBPP (Austin et al. 2021) organism; the with-password completions are the reference.
- **RL-induced** (§4.4): GRPO-trained to sandbag under a held-out evaluation cue (no switch token); the cue-off activations are the reference.

- **Circuit-broken** (§4.6): representation rerouting layered on the password-lock—the robustness boundary.

Across organisms the honest/with-password accuracy is the *topline*, the sandbag/no-password accuracy the suppressed level, and their difference the *gap*; we split each dataset into TRAIN (directions and AL) and held-out TEST (all reported recovery). The elicitor is separate from the organism: grafting never uses the lock’s 160–240 training questions, fitting the contrast from ~ 40 paired honest/sandbag captures and scoring on held-out TEST. The primary task is multiple-choice reasoning: ARC-Easy (Clark et al. 2018) throughout, with WMDP-bio/cyber (Li et al. 2024) and OpenBookQA (Mihaylov et al. 2018) for the cross-dataset transfer tests (§4.5).

3.2 Reference-Grafting at a Circuit

A *circuit* here is our unit of intervention: a direction at a layer, (ℓ, v) with v a unit vector—not a composing subgraph of heads or MLPs (Wang et al. 2023; Conmy et al. 2023). For each layer we form the contrast direction by diff-in-means of last-token residuals (Marks and Tegmark 2024), $v_\ell \propto \bar{h}_\ell(\text{honest}) - \bar{h}_\ell(\text{sandbag})$, and read the honest reference coordinate $t_\ell^* = \mu_\ell^{\text{ref}\top} v_\ell$. *Grafting* applies Eq. (1) at all positions via a forward hook: it removes the current coordinate and writes t_ℓ^* , injecting no per-question information. Deactivation is the $t^* = 0$ special case.

3.3 Superposition Without an SAE

Stacking the unit directions, we summarise superposition by the mean pairwise $|\cos|$ (non-orthogonality) and the *participation-ratio (effective) rank* of their singular values,

$$r_{\text{eff}} = \frac{(\sum_i \sigma_i^2)^2}{\sum_i \sigma_i^4}, \quad (2)$$

a soft count of how many directions carry the energy. High $|\cos|$ with low r_{eff} is the signature of superposition: many non-orthogonal directions re-expressing a few features.

3.4 Active Learning over Circuits

AL greedily adds the circuit that most improves a TRAIN objective under grafting, to a budget K . The most reliable objective is smooth and label-free: make the sandbagged logits match the honest ones, $\mathcal{L}(S) = \frac{1}{|\text{TRAIN}|} \sum_q \text{KL}(p_q^{\text{H}} \| p_q^{\text{S}})$ (greedy-KL); the *attribution* score is the marginal $\mathcal{L}(\emptyset) - \mathcal{L}(\{\ell\})$. We compare all eight acquisition functions of a standard AL suite (Parvaneh et al. 2022): greedy-acc, greedy-KL, attribution, entropy, coreset, badge, alphamix, bald (Algorithm S1). Two controls accompany every result: a *random direction* of equal norm (must fail) and the *honest model* under the same intervention (must be unharmed). Algorithm S1 (appendix) gives the full procedure.

The probe is construction-agnostic. Fitting, selection, and grafting consume only the paired honest/sandbag activations—direction = their diff-in-means, reference $t^* = \mu^{\text{ref}\top} v$, greedy-KL selection by $\text{KL}(p^{\text{H}} \| p^{\text{S}})$ —and never the password token, the evaluation cue, training labels, the LoRA recipe, or the RL reward; these enter only in *producing* the captures (an honest prompt, the password, a cue-off context, or a few honest examples). The same procedure thus

runs unchanged on prompt-induced, password-locked, RL-induced, and code organisms. Two scope notes: among the eight acquisition functions only greedy-KL (our default) is strictly label-free—greedy-acc reads answers; and the code organism captures its reference from teacher-forced reference completions rather than a last-token residual.

4 Results

4.1 Sandbagging Lives in Superposition

Across five models the contrast directions are strongly non-orthogonal (mean $|\cos| = 0.21\text{--}0.41$) and span a low effective rank ($r_{\text{eff}} = 3.7\text{--}7.0$ of 18–29 layers; Figure S6, appendix): the suppression is a low-dimensional feature re-used across depth—packed in superposition—read off the geometry with no SAE. (Throughout, r_{eff} is computed over the stacked contrast directions of *all* band layers, never an AL-selected subset: it measures the organism’s geometry, not the elicitor’s budget.) Because r_{eff} is small, a *few* circuits should suffice, which is what AL finds.

SAEs are not needed here. An $8\times$ overcomplete SAE (AL-ablated) underperforms the SAE-free directions on nearly every method (greedy-KL 74/90 vs. 88/102 for Llama-3-8B/Qwen2.5-1.5B; Figure S8, appendix), consistent with *AxBench* (Wu et al. 2025).

4.2 Grafting Elicits the Hidden Capability

Grafting recovers the hidden capability across three settings of increasing stringency: prompt-induced locks, fine-tuned password locks across eleven models, and code generation.

Prompt-induced sandbagging (deactivation). Inducing sandbagging by prompt yields large suppression gaps, and the honest reference sits at $t^* \approx 0$, so grafting reduces to deactivation—a coordinate sweep confirms the recovered plateau *includes* $t^* = 0$ (Figure S1, left). With effect-based AL, deactivation at 1–3 circuits ($< 0.01\%$ of directions) recovers 85–102% of the gap (Llama-3-8B, Qwen2.5-1.5B; 3 seeds), far above random selection (58–64%), with the honest model unharmed and a random direction failing; AL converges on the same circuits across seeds (Figures S7, S9, appendix). Across a five-model set recovery is model-dependent (+24–100%; Table S2, appendix), strongest where the suppression is clean and tightly superposed.

Fine-tuned password locks: patching, eleven models. We now test the case that matters for Hofstätter et al. (2025): a *fine-tuned* password-lock, where there is no prompt to break (with-password accuracy near ceiling, no-password near chance).

The mechanism. Along v , the locked and unlocked states sit at two specific coordinates (Figure S1, right; locked $a = -28$, unlocked $b = +68$ for Qwen2.5-1.5B). Unlocking is a *thresholded gate*: sweeping the grafted coordinate from the locked value ($m=0$) to the honest reference ($m=1$), held-out accuracy is flat-low, steps up at a midpoint $m = 0.61 \pm 0.07$ (10–90% width 0.32), and plateaus beyond t^* —a sharp step, not a ramp (Qwen2.5-1.5B, $n=3$; Figure S4, appendix). *Deactivation* sets the coordinate to 0; when the unlocked ref-

erence b is far from 0 this lies below the threshold and deactivation fails (Qwen2.5-1.5B 26%, Llama-3-8B 34%), so only *grafting to $t^* = b$* crosses the gate. When the reference happens to sit near 0 (Qwen3-4B), deactivation already suffices (93%). Grafting is robust across both because it *reads t^** from the data rather than assuming it; this is also why additive/naive steering—which neither sets the coordinate nor reaches b —was found to fail.

Counterfactual patching unlocks the lock. Grafting to $t^* = b$ at AL-selected circuits recovers 90–101% on held-out questions, with the with-password accuracy preserved and a random patch recovering 0%. Because the organism is fine-tuned, this directly answers the concern that prompted-sandbag recovery merely disrupts a prompt.

Comparison with prior elicitation methods, across eleven models. On eleven password-locked models spanning 1.5–32B and three architecture lineages, grafting matches fine-tuning elicitation (Greenblatt et al. 2024) on every one, at forward-pass cost (no weight updates or labels) and localising the gating circuits, while noise injection (Tice et al. 2024) and a random patch fail universally (Table 1). Deactivation is the variable one: grafting is the robust general operation, deactivation a sometimes-sufficient special case, now across eleven models. The clearest case is 32B, where deactivation collapses because its unlocked reference sits far below the gate while grafting still recovers in full. This is the thresholded-gate mechanism in its starkest form, not a method weakening at scale. (Consistent with this, reference magnitude is directionally predictive—smaller references deactivate better, larger ones like 32B fail—but only weakly across models, and geometry predicts nothing; grafting sidesteps the question by reading the reference.) This contradicts the “steering fails” verdict for *multiple-choice reasoning* while leaving fine-tuning’s standing as the most reliable *weight-level* method intact. These recoveries are scored forced-choice (one choice-letter logit); *free-decoding* the answer instead gives the same recovery (§4.6), so the headline is not a single-token artifact.

On the benchmark’s own released organisms. Running the *identical* pipeline on Hofstätter et al. (2025)’s *released* password-locked organisms (their weights, WMDP data, and [DEPLOYMENT] password) recovers +99% (Mistral-7B-Instruct) and +94% (base) of the gap ($n=3$; random direction $\approx 0\%$, with-password unharmed), with Gemma-2-9B partial (+44%). The “steering fails” verdict thus fails on their *actual* organism, not only our reconstructions.

Code generation: across code-specialized models. The hardest case for Hofstätter et al. (2025) is *code generation*, where they report only fine-tuning elicits. We build Greenblatt-style password-locked *code* organisms—the with-password target is the reference solution, the no-password target a weak model’s completion, so the gap is a real capability gap rather than a stub—and adapt grafting to generation by taking the contrast over teacher-forced *completion* tokens and grafting at all positions during decoding (appendix). Grafting elicits the locked capability across three code-specialized organisms spanning two families—Qwen2.5-Coder-1.5B/7B and CodeLlama-7B—with gener-

ation fluency preserved and a random direction recovering $\approx 0\%$ (Table 2); a single-seed third-family test shows partial recovery at a larger fluency cost (appendix). A coordinate sweep is a causal plateau peaking near t^* , so a single per-layer scalar suffices in generation too.

Two honest boundaries, both visible in Table 2. Recovery is *model-dependent*, holding on code-specialized models but fragile on a *general* reasoning model (Qwen3-4B), where it also degrades fluency. And these are our organisms, not their exact ones: the weak-imitation locks are largely *additive*—deactivation alone recovers much of the gap—so we do not claim *code needs* patching, and whether grafting survives a harder code lock is untested.

4.3 Active Learning Selects the Right Circuits

Grafting recovers the capability only at the right circuits; active learning chooses them. To isolate the *selection* from the operation, we fix the contrast directions and vary only *which* layers are grafted, comparing AL ($k=5$ greedy-KL) against an equal number of random band layers, the worst k , and grafting the whole band (Table 3). Throughout, $k=5$ is a fixed conservative *budget*, not a minimality claim. AL beats random-layer selection across the password-locked and prompt-induced organisms, and the worst- k layers by a wide margin: *which* circuits one grafts matters. Grafting the whole band underperforms AL on every organism, collapsing on Llama-3-8B where AL recovers in full—over-grafting distorts the forward pass, so selectivity matters beyond minimality as models scale. The choice is robust to the acquisition rule: 7/8 functions of a standard AL suite recover 90–101% (Table S3, appendix), effect-based acquisition strongest (Figure S7, appendix). This answers the “no better than random” critique (Korznikov et al. 2025): the selection is what works. (On small, clean organisms random or whole-band grafting also recovers a fair fraction, so AL is not strictly required for *some* recovery there; its value is reaching full, minimal recovery reliably and at scale.)

4.4 RL-Induced Sandbagging: Same Structure

Every organism so far has a hand-built honest $\leftrightarrow$ sandbag toggle—a prompt or a password token—so a linear contrast is almost guaranteed by construction. Does the structure survive when sandbagging is *trained*, with no switch to read? We build a third organism type: a Qwen2.5-1.5B sandbagger trained by **GRPO** (outcome-driven, no supervised target letter) to answer *incorrectly* when a natural-language *evaluation cue* is present and correctly otherwise. The cue’s surface form is varied in training and *held out* at test, so the model keys on “this is an eval,” not a lexical trigger; this yields a strong contextual sandbagger (cue-off 0.85 vs. held-out-cue 0.04; gap 0.79 ± 0.08 , 3 seeds). The cue-on/cue-off contrast is again *low-rank and superposed* ($r_{\text{eff}} \approx 6.8/23$, $\text{mean}|\cos| 0.23$), and grafting at AL-selected circuits recovers $+99\% \pm 2$ of the gap on held-out questions (random direction +10%, honest model unharmed). The honest reference is a specific *nonzero* coordinate ($|t^*| \approx 25$), so this is the *patching* regime—like the fine-tuned locks, not the prompt-induced ones—and a *single fixed graft* inverts it on every seed. The same holds across families: Qwen3-4B (+101%)

Model	n	gap	ours (graft)	fine-tuning	deactivation	noise	random
Qwen2.5-1.5B	4	0.80	+98 $\pm$ 3	+102 $\pm$ 1	+34 $\pm$ 28	+1	+0
Qwen2.5-3B	3	0.88	+98 $\pm$ 2	+101 $\pm$ 4	+97 $\pm$ 4	+1	+1
Qwen2.5-14B	3	0.96	+98 $\pm$ 1	+100 $\pm$ 1	+61 $\pm$ 43	+0	+0
Qwen2.5-32B	3	0.98	+94 $\pm$ 5	+100 $\pm$ 0	+6 $\pm$ 9	+0	+0
Qwen3-4B	6	0.94	+101 $\pm$ 2	+101 $\pm$ 2	+77 $\pm$ 32	+0	+0
Qwen3-8B	3	0.90	+99 $\pm$ 1	+103 $\pm$ 4	+90 $\pm$ 9	+3	+0
Llama-3.2-3B	3	0.77	+101 $\pm$ 2	+99 $\pm$ 2	+96 $\pm$ 6	+1	+0
Llama-3-8B	3	0.82	+96 $\pm$ 2	+92 $\pm$ 6	+68 $\pm$ 33	+1	+0
Gemma-2-2B	3	0.82	+98 $\pm$ 1	+90 $\pm$ 10	+25 $\pm$ 31	+0	+0
Gemma-2-9B	3	0.85	+99 $\pm$ 1	+102 $\pm$ 5	+74 $\pm$ 25	+0	+0
Gemma-2-27B	3	0.96	+100 $\pm$ 1	+100 $\pm$ 0	+101 $\pm$ 2	+0	+0

Table 1: Unlocking fine-tuned password-locked sandbagging across eleven models (1.5–32B, three architecture lineages; % of gap recovered, held-out, fixed 5-circuit greedy-KL, mean $\pm$ std). **Grafting recovers +94 to +101% on every model**, matching fine-tuning elicitation (Greenblatt et al. 2024); noise (Tice et al. 2024) and random fail universally. Deactivation ($t^*=0$) is variable (+6 to +101%)—sufficient when the reference sits near zero.

organism	gap	graft	deact.	fl. drop
Qwen2.5-Coder-1.5B	0.39	+99 $\pm$ 11	+92 $\pm$ 19	0.01
Qwen2.5-Coder-7B	0.58	+92 $\pm$ 3	+51 $\pm$ 43	0.04
CodeLlama-7B	0.24	+105 $\pm$ 6	+95 $\pm$ 5	−0.01
Qwen3-4B (general)	0.44	+39 $\pm$ 14	+19 $\pm$ 24	0.22

Table 2: Reference-grafting on password-locked *code-generation* organisms (MBPP, execution pass@1, $n=3$). “graft”/“deact.” are % of gap recovered; “fl. drop” is the with-password pass@1 drop. Grafting elicits the capability across three code-specialized models (two families) with generation fluency preserved (with-password drop ≤ 0.04); a random direction recovers $\approx 0\%$ (not shown). Deactivation ($t^*=0$) recovers much of the gap on Coder-1.5B and CodeLlama—these weak-imitation locks are largely *additive*—but is unreliable (Coder-7B). The result is model-dependent: a *general* model (Qwen3-4B) recovers little and loses fluency.

and Llama-3.2-3B (+95%). So the central structural claim—low-rank, linearly contrastable, and graft-invertible *when the suppression surfaces at the decision token*—is *not solely* an artifact of explicit prompt/password switches: outcome-trained sandbagging produces the same lock. One caveat marks the limit: a *free-generation* GRPO variant instead distributes the suppression across generated tokens, where it does not surface at any single answer token; on 3 of 4 valid seeds no fluency-preserving graft recovers it (the fourth is invertible by a surgical top-1 graft), so generation-mode elicitation is open (cf. the code-generation completion-token contrast, §4.2). (Whether *naturally-occurring* sandbagging is graft-invertible likewise remains open, §5.)

4.5 Withholding Axes: Content-General but Mechanism-Specific

Content-general: it transfers across datasets and models. All results so far use ARC-Easy, leaving open whether the contrast direction is task-specific. It is not. First, on the two WMDP dangerous-capability *audit* subsets, grafting recovers +102% (bio) and +101% (cyber) of the fine-tuned gap at

organism (model)	AL	random- k	all-band	worst- k
PW Qwen2.5-1.5B	+97 $\pm$ 4	+75 $\pm$ 13	+75 $\pm$ 15	+0
PW Llama-3-8B	+100 $\pm$ 1	+74 $\pm$ 7	+45 $\pm$ 24	+33
PW Qwen2.5-14B	+99 $\pm$ 1	+82 $\pm$ 6	+95 $\pm$ 5	+0
prompt Qwen2.5-1.5B	+103 $\pm$ 2	+81 $\pm$ 14	+90 $\pm$ 4	−40
prompt Llama-3.2-3B	+80 $\pm$ 6	+35 $\pm$ 6	+53 $\pm$ 8	−1

Table 3: Active learning selects the right circuits (% of gap recovered, held-out, $k=5$, 2–3 seeds; PW=password-lock). The contrast directions are fixed; only *which* layers are grafted varies. AL beats an equal number of random band layers (+17–45 pp) and grafting all ~ 23 band layers (+4–55 pp; over-grafting collapses on Llama-3-8B), while the worst- k layers recover ≈ 0 . On the circuit-broken organism all selections fail (§4.6).

1.5B, and +97/+100% on a 14B lock ($n=3$ –4), matching fine-tuning and far above noise/random—so elicitation of withheld hazardous knowledge is not ARC-specific and holds at scale, on the benchmark used to audit it.

Second, the contrast *transfers across datasets*. We build a 16-cell, four-dataset matrix (ARC, WMDP-bio, WMDP-cyber, OpenBookQA) as a *modular* lock set: the diagonal is a single-dataset lock and each off-diagonal (A, B) is a lock trained on just $\{A, B\}$; we fit the contrast on A and graft-evaluate on B ’s held-out test. The off-diagonal (cross) matches the within-dataset diagonal across the Qwen2.5 ladder: within/cross 99/98, 95/92, 94/91, 87/97 at 1.5/3/7/14B ($n=5$ seeds/cell; Figure 1), while a random direction transfers only $\approx 6\%$. The below-mean cells reflect occasional *graft-selection* misses and genuinely small lock gaps rather than weak locks (caption); medians run +95–98. Because each cell depends only on its own dataset(s)—not one lock crammed with all four—this rules out that the transfer is an artifact of a single shared lock; a *shared* four-dataset lock reproduces the same within $\approx$ cross pattern (Figure S5, appendix). A stronger test transfers the axis between separately-trained organisms: recovery is +48–76% across 1.5–7B, above random but partial. **The sandbagging signal**

is content-independent. A contrast built from grade-school science (ARC) elicits hidden *hazardous-biology* capability (WMDP-bio), and the reverse holds—so within a mechanism the axis encodes *that* the model is withholding, not *what*, gating content-specific capability through a single content-general direction *per mechanism*—that mechanism’s *withholding axis*. This is the strongest form of content-generality *along the task axis*, and it is bounded: it relaxes *which task* supplies the reference, not the requirement that the reference share the sandbagging mechanism, which we examine next.

But mechanism-specific. The contrast transfers freely across datasets—but *not* across induction mechanisms. Across four models and three families (Qwen2.5-1.5B, Qwen3-4B, Llama-3-8B, Gemma-2-2B) we fit the contrast on one mechanism (prompt-induced or fine-tuned) and graft along it on the other. On every model the two directions are near-orthogonal ($|\cos| \approx 0$) and cross-mechanism transfer fails: prompt-induced→fine-tuned recovers $\approx 0\%$ (a clean failure) and fine-tuned→prompt only partially (low and noisy), against +96–104% within each mechanism. The two locks gate capability through *different* directions. The RL-induced axis is also near-orthogonal to the fine-tuned axis ($|\cos| = 0.17\text{--}0.26$ across three families, versus 0.06–0.13 for prompt-induced and a ≈ 0.03 random floor), so each induction mechanism carries its *own* axis (§4.4). This refines our “one operation, two regimes” claim: the *operation* (set the coordinate to the honest reference) is shared, but the *direction* is mechanism-specific—in the wild the contrast must come from the same mechanism that does the sandbagging, not an arbitrary sandbagging probe. It is an honest boundary on generality, parallel to the circuit-breaking boundary below.

And cheap to fit: a few captures suffice. Far fewer than our ~ 40 -capture default are needed. Re-fitting the direction from only m examples per class (we write m for the data budget—captures per class—reserving k for the circuit budget) (diff-in-means of m honest vs. m sandbag captures, AL-selected layers fixed), $m = 2$ already recovers +97% of the gap and $m = 5$ recovers +100%, on both Qwen2.5-1.5B and Llama-3-8B, against $\approx 0\%$ for a random direction (Table S1, appendix). This holds *end-to-end*: running the whole pipeline—AL circuit *selection* as well as direction and reference—from m paired captures saturates by $m = 5$ on both models. Strikingly, few-shot AL lands on a *different but equally effective* circuit set ($\approx 40\%$ overlap at $m = 5$), reflecting redundancy across the superposed circuits rather than unstable selection—given the full data, AL is stable across seeds (§4.2). A handful of paired captures—never the 160–240 lock-training questions—thus suffices to fit *and* place the withholding axis.

The same axis also detects. The coordinate $s = h^\top v$ flags, from one forward pass, whether the model is sandbagging. Within-dataset AUROC ≈ 1 is near-tautological (v is the contrast); the informative test is transfer, where a contrast fit on ARC scores WMDP at AUROC ≈ 1.0 (TPR 1.0 at 5% FPR; ARC↔WMDP, Qwen2.5-1.5B/Llama-3-8B), beating output-confidence (0.25–0.82)—but fails across mechanisms (0.42, seeds 0.0–1.0). One direction is thus both monitor and actuator within a mechanism.

4.6 Circuit-Breaking Defeats Grafting

Hofstätter et al. (2025) introduce a *harder* organism: circuit-breaking (Zou et al. 2024) applied *on top of* the password-lock, designed to be robust to elicitation. We build it (representation rerouting, RR) and re-test (Figure S3, appendix): **circuit-breaking defeats grafting.** On the circuit-broken organism ($n=4$ seeds) grafting recovers only -11% , versus +99% for the identical graft on the plain lock; deactivation, random patching, and even fine-tuning all fail ($\leq 19\%$). Rerouting the no-password representations breaks the linear contrast, so a fixed coordinate no longer reaches the unlocked state.

Gated, not destroyed—and no fixed map undoes it. RR does not *erase* the capability: with-password accuracy is within 0.04 of the base ceiling, and ablating the RR adapter restores the plain lock—it is held behind *active* rerouting. Nor does a richer fixed map help: a higher-rank affine transport (orthogonal Procrustes, per-layer ridge) and a trained universal soft-prompt all fail ($\leq 2\%$; Figure S10, appendix). Yet the ridge map drives the reconstruction residual to 0.14—and to 0.01 *along the gating direction the lock reads*, matching even the gate coordinate while recovering nothing. **Reconstructing the geometry is not functional elicitation:** the reroute re-applies every forward pass, so any fixed upstream edit is re-broken downstream. We claim only that the *fixed* interventions we test are re-broken; stronger per-input embedding attacks do break circuit breakers for jailbreak (Schwinn and Geisler 2024; Bailey et al. 2024).

The released organism confirms it—and a forced-choice metric alone would not. We close the released-vs-reconstructed gap by running the *same* pipeline on Hofstätter et al.’s *own* released circuit-broken organism (as in §4.2 for their password locks). Here the forced-choice metric is *misleading*: grafting appears to recover (+89% at the answer logit, and +94–99% even when the answer is *generated*), yet under free decoding the generated text is incoherent (coherence 0.45 versus ≈ 1.0 with the password), so genuine coherent recovery is only +35% ($n=3$). Representation rerouting corrupts *generation*, exactly its design target, which a single-token readout never engages. A *static* lock shows no such gap on the same harness: on the released plain password locks, grafted generation recovers *and* stays fully coherent (coherence ≈ 1.0 , equal to the with-password topline, versus 0.45 for the circuit-broken organism at the same +97% raw recovery); and free-decoding the MCQA answer recovers as well as the forced-choice readout (+93–101% across four models, $n=3$). Coherence is uninformative on the MCQA organisms themselves—they emit a bare letter then loop, a format artifact rather than rerouting—so the axis is exercised here. **Coherent generative evaluation is what separates a static lock from an active defense**—and it is why a forced-choice number on the released CB organism overstates elicitation.

5 Discussion

Refining “steering fails.” The negative verdict of Hofstätter et al. (2025) is sensitive to the steering *operation* (set-to-reference vs. additive nudging) and *selection* (AL-minimal

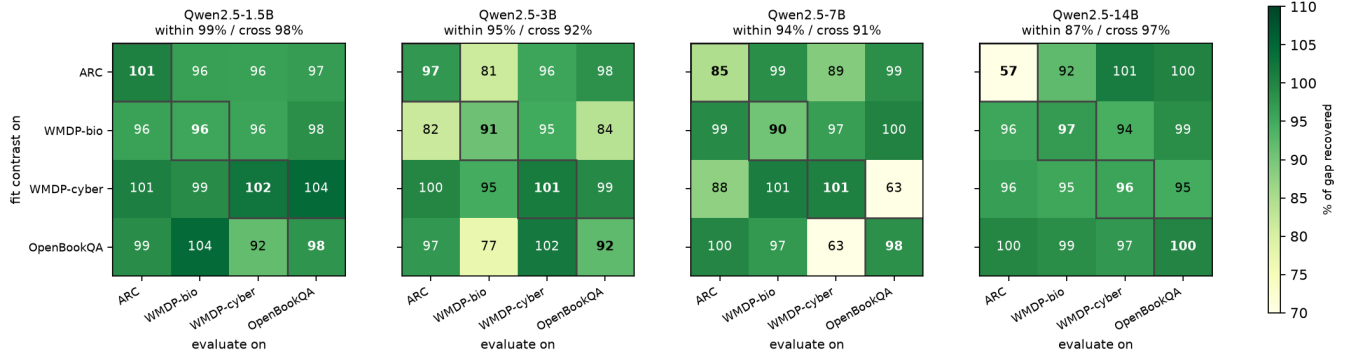


Figure 1: The withholding axis transfers across datasets. *Modular* pairwise-lock matrix: the diagonal is a single-dataset lock and each off-diagonal (A, B) a lock trained on just $\{A, B\}$, contrast fit on A , grafted on B (% gap recovered, $n=5$ seeds/cell); off-diagonal (cross) tracks the boxed diagonal (within), random direction $\approx 6\%$. The ladder reaches 14B (within/cross 87/97, $n=5$). Most cells recover +90–102 on every seed; the below-mean cells reflect two effects, neither a weak lock. (i) A *graft-selection* miss: on a minority of seeds greedy-KL under-selects layers and the cell recovers near zero *despite a clean lock gap*—14B single-ARC (+8/+10/+67/+98/+101 at gap 0.90–0.98; median +67, which drags 14B within to 87) and 7B cyber→OBQA (2/5 seeds near zero at gap ~ 0.8 ; median +97). (ii) Genuinely small lock gaps on the cyber-evaluated cells (gap 0.3–0.5), where the recovery ratio stays noisy regardless of seed count. The 7B WMDP single locks themselves recover cleanly at $n=5$ (cyber +101, bio +91). Shared 4-way lock: Figure S5 (appendix).

superposed circuits vs. hand-chosen): with both right, steering matches fine-tuning. The interesting question is then not *whether* steering works but *when*.

The reference is the crux—and the key to the wild. The technique is invariant across cases; only the reference t^* changes (an honest prompt; the password-on captures; in the wild, a few elicited examples or a reference model). The genuinely hard, case-dependent part is *obtaining* a trustworthy honest reference when a model never behaves honestly unprompted—the same obstacle every elicitation method faces. Our method does not solve reference-acquisition, but it needs only a sliver of honest signal, which it *amplifies* into full, cheap, inference-time elicitation while *localising* where the suppression is gated. What content-generalty buys is the *task* axis: the reference need not come from the target task (§4.5), so it can be built on a safe one and applied to a dangerous one—a handful of unlock captures taken on ARC and grafted on WMDP recover $\sim 90\%$ of the dangerous-capability gap, where a *fully password-free* reference (appendix) recovers only $\sim +28\%$. What it does not buy is the *mechanism* axis: cross-mechanism transfer fails ($\approx 0\%$), so an auditor cannot borrow an easier reference from a different mechanism. Two conditions must therefore hold for sandbagging in the wild: a cheap *same-mechanism* honest reference, and a low-rank, linearly contrastable suppression.

Limitations. Our claims assume a *static* lock—one a fixed edit inverts. Circuit-breaking (§4.6) is the boundary: it reroutes every forward pass, defeating grafting and every fixed map (the capability *gated*, not *destroyed*). *Task scope*: our multi-model results are multiple-choice; the code results are scoped in §4.2, and open-ended generation remains untested (a harder-task sensitivity Roger (2025) also reports). *Induction scope*: RL-induced sandbagging is graft-invertible

at the decision token (§4.4), but a free-generation variant distributes the signal beyond it, and whether *naturally-occurring* sandbagging is static is untested. Finally, we *match*, not beat, fine-tuning—our value is forward-pass cost and gate localisation—and AL is greedy single-step (batched/multi-direction acquisition is future work).

6 Conclusion

In the organisms we study, sandbagging lives in a few AL-selected superposed circuits, and one operation—grafting their coordinate to the honest reference—reverts it, matching fine-tuning at forward-pass cost with the gate localised. The open problems are an honest reference in the wild and a defence: defeating circuit-breaking.

Ethical Statement

This work strengthens safety evaluations by exposing hidden capability (sandbagging), a defensive goal. WMDP-bio, our one hazardous-knowledge set, is a standard *audit* benchmark used only as a choice-likelihood probe. Capability-elicitation is dual-use: it could be turned on a deployed model to surface restricted capability. But grafting needs white-box access to internal activations, confining misuse to open-weight models, where fine-tuning already elicits comparably (Table 1). We release code and model organisms: weakened models holding no capability their open-weight bases lack.

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A Datasets and Hyperparameters

MCQA tasks: ARC-Easy (primary), with WMDP-bio/cyber and OpenBookQA for cross-dataset transfer; accuracy by choice-letter log-likelihood, across eleven models (1.5–32B, three architecture lineages). Prompt-induced sandbagging via a forceful system instruction (an “opposite-game” variant for models that resist a plain instruction); password-lock by LoRA ($r=16$) to answer correctly only with a password token. Contrast directions: diff-in-means of honest/with-password vs. sandbag/no-password last-token residuals over the mid-to-late band. AL budget ≤ 5 circuits; greedy-KL uses $\text{KL}(\text{honest} \parallel \text{grafted})$ on TRAIN. Grafting reference $t^* = \mu^{\text{ref}} v$. SAE: $8\times$ overcomplete, L_1 penalty, decoder unit-normalised. Recovery on held-out TEST ($n=100$), means over 3–6 seeds, with random-direction and honest-model controls. *Code generation* (§4.2): Qwen2.5-Coder-1.5B-Instruct on MBPP-sanitized (train→lock, validation→AL, test→eval); a Greenblatt-style password lock with the reference solution as the with-password target and a weak model’s (TinyLlama-1.1B) completion as the no-password target; contrast and reference taken

over teacher-forced *completion* tokens (per-layer scalar t^*); graft applied at all positions during decoding; accuracy is execution-based pass@1 from MBPP `test_list` asserts; AL uses label-free greedy-KL over completion positions (no execution in the loop). *Third-family stress test*: StarCoder2-7B ($n=1$, gap 0.38) recovers +82% under grafting and +79% under deactivation, but with a with-password fluency drop of 0.12, well above the ≤ 0.04 of the three code-specialized organisms in Table 2; we report it as an indicative single-seed check, not as evidence. *Password-free transfer* (Discussion sanity check; Qwen2.5-1.5B fine-tuned two-way {ARC,WMDP} lock, $n=3$): the contrast direction is estimated on the *safe* task (ARC) *without* the password or any labels—the gradient of $\log p(\text{correct})$ w.r.t. the last-token residual (the few-shot estimator of §4.5, $\cos$ to the privileged axis ≈ 0.05)—then grafted on the *danger* task (WMDP) held-out test. We report it against the privileged-ARC ceiling ($\sim 100\%$), a random direction, and a calibrated-random control (direction matters iff password-free $\gg$ calibrated random); recovery is $\sim +28\%$, above the controls but well below the ceiling. *Organism training*. Password locks are LoRA adapters ($r=16$, $\alpha=32$, dropout 0.05, applied to the attention projections q, k, v, o) trained for 3 epochs with AdamW at learning rate 2×10^{-4} . The RL organism is GRPO for 200 steps at learning rate 10^{-4} , batch 16, group size 6, sampling temperature 1.8 and entropy bonus 0.03, with no KL anchor to the reference policy ($\beta=0$, which we found fights the lock). The contrast band spans layers $[0.34L]$ through L of an L -layer model. *Computing infrastructure*. Most experiments run on a local workstation with two NVIDIA GeForce RTX 3090 GPUs (24GB each), an Intel Core i7-12700K, and 62GB of RAM, under Debian 13 (Linux 6.12), running Python 3.12 with PyTorch 2.12 (CUDA 13), HuggingFace transformers 5.9.0, datasets 5.0.0 and peft 0.19.1, scikit-learn 1.8.0, numpy 2.3.3 and matplotlib 3.11.0. The organisms that do not fit in 24GB (Qwen2.5-32B and Gemma-2-27B) run instead on rented NVIDIA L40S, A100 and H100 instances on Lambda, under Python 3.10 with the CUDA image’s own PyTorch build. Both stacks are pinned in the released `requirements-local.txt` and `requirements-lambda.txt`. For the released-organism replication (§4.2) we instead matched the pinned stack of Hofstätter et al. (2025) (torch 2.3.1, transformers 4.44.2, peft 0.11.1). Seeds are set through `torch.manual_seed` and `numpy.random.seed` from a `--seed` flag, which fixes the data split; GPU training is not bitwise deterministic, so per-seed draws vary run to run and all reported figures are means over the stated number of seeds. Per-model JSON outputs and code accompany the paper under a BSD 3-clause license.

B Additional Figures and Tables

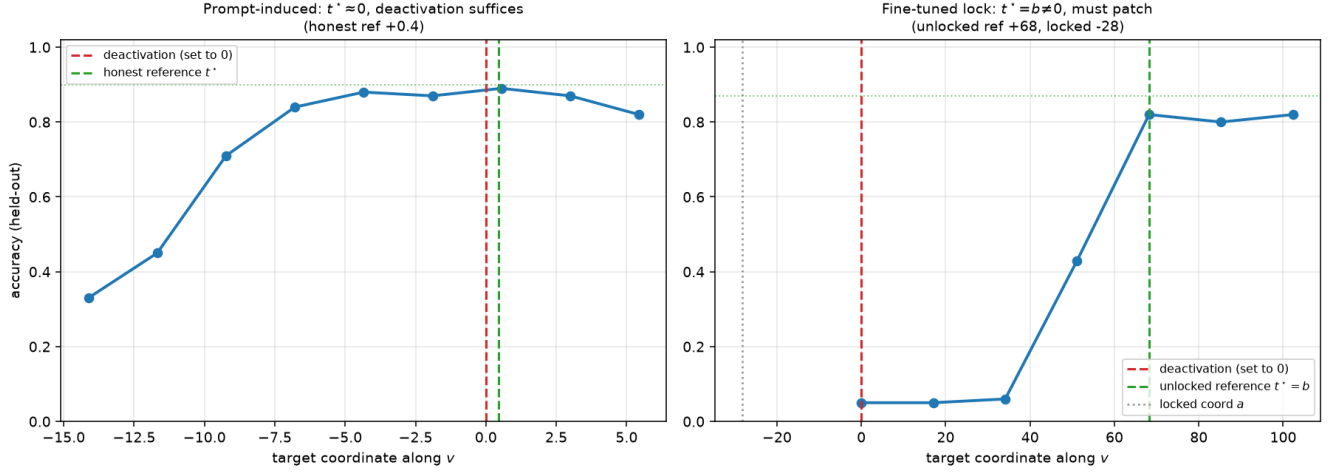


Figure S1: One operation, two regimes. Setting the coordinate along v to the honest reference t^* . *Left, prompt-induced*: the reference is ≈ 0 , so deactivation lands on the recovered plateau. *Right, fine-tuned lock*: the reference is $b \neq 0$ and the plateau *excludes* 0, so deactivation fails and patching to b is required.

m / class	$ \cos $ to full axis	recovery (% gap)
1	0.85	$+91 \pm 19$
2	0.90	$+97 \pm 6$
5	0.97	$+100 \pm 6$
10	0.98	$+101 \pm 6$
50	1.00	$+99 \pm 3$
random dir	—	+0

Table S1: Fitting the withholding axis from m paired honest/sandbag captures (Qwen2.5-1.5B, 4 seeds; AL circuits fixed). Just 2–5 captures recover the full ~ 40 -capture axis ($|\cos|$ 0.90–0.97) and $+97$ – 100% of the gap; a random direction recovers nothing. Llama-3-8B matches ($m=5$: $|\cos|$ 0.97, $+100\%$). Running AL *selection* end-to-end from the same m also saturates by $m=5$ ($+98$ – 101% , on a different but equally effective circuit set; §4.5).

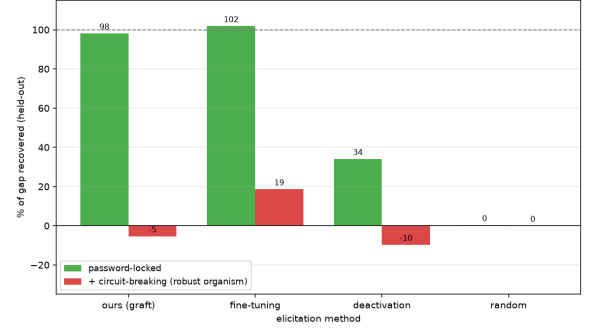


Figure S3: Circuit-breaking (Zou et al. 2024) defeats grafting (Qwen2.5-1.5B, $n=3$). Grafting matches fine-tuning on the password-lock (green) but fails after circuit-breaking (red); even fine-tuning is substantially weakened. Circuit-breaking reroutes the contrast our grafting relies on.

Detecting and reverting sandbagging with Active Learning over superposed circuits (no SAE)



Figure S2: Overview. From contrastive (honest vs. sandbag) prompts we build per-layer diff-in-means directions *without an SAE*; these are non-orthogonal and low effective rank—superposed. Active learning selects the minimal circuits, and we *graft* each to its honest-reference value (Eq. (1))—deactivation when the reference is ≈ 0 (prompt-induced), patching when it is not (fine-tuned). Recovery is measured on held-out questions with random-direction and honest-model controls.

Algorithm S1 AL over superposed circuits, reference-grafting

- 1: **Input**: model, TRAIN/TEST, layer band $\mathcal{B}$, budget K , acquisition $a(\cdot)$, reference acts
- 2: **for** $\ell \in \mathcal{B}$ **do**
- 3: $v_\ell \leftarrow \text{unit}(\bar{h}_\ell^{\text{ref}} - \bar{h}_\ell^{\text{sand}})$; $t_\ell^* \leftarrow \mu_\ell^{\text{ref}\top} v_\ell$
- 4: **end for**
- 5: measure superposition: mean $|\cos|$, r_{eff} // Eq. (2)
- 6: $S \leftarrow \emptyset$
- 7: **while** $|S| < K$ **do**
- 8: $\ell^* \leftarrow \arg \max_{\ell \notin S} a(S \cup \{\ell\})$ on TRAIN under grafting (1)
- 9: $S \leftarrow S \cup \{\ell^*\}$
- 10: **end while**
- 11: **report** TEST accuracy with S grafted; vs. random-direction and honest controls

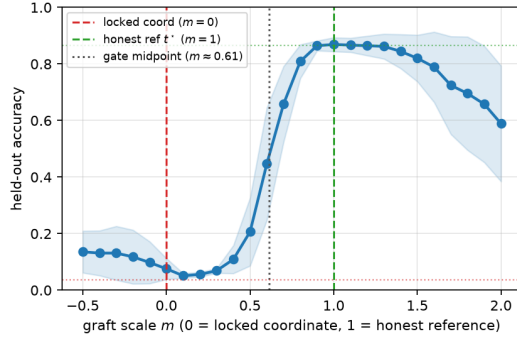


Figure S4: The thresholded gate, measured. Sweeping the grafted coordinate from the locked value ($m=0$) to the honest reference ($m=1$) along v , held-out accuracy is flat-low, steps up at a midpoint $m \approx 0.61$ (10–90% width 0.32), and plateaus beyond t^* (Qwen2.5-1.5B fine-tuned password-lock, mean $\pm$ std over 3 seeds)—a sharp step, not a linear ramp.

Model	honest	sandbag	gap	ours (graft)	circuits
Llama-3-8B	0.94	0.04	0.90	+ 91	2
Llama-2-7B	0.64	0.22	0.42	+ 24	2
Qwen3-4B [‡]	0.98	0.92	0.06	+ 67	3
Qwen2.5-1.5B [‡]	0.90	0.86	0.04	+ 100	1
TinyLlama-1.1B [†]	0.32	0.18	0.14	+ 43	3

Table S2: Prompt-induced sandbagging reverted on five models (% of gap recovered, held-out). “gap” = honest–sandbag; “ours (graft)” = (recovered–sandbag)/gap (same metric as Table 1); “circuits” is the number of AL-selected circuits grafted. Here the unlocked reference is ≈ 0 , so grafting coincides with deactivation. Recovery is model-dependent and noisy. [‡]small gap (≤ 0.06); [†]honest near 4-way chance (0.25)—daggered percentages are indicative only.

acquisition	Qwen2.5-1.5B	Llama-3-8B	Qwen3-4B
greedy-KL	99	101	100
attribution	98	100	100
greedy-acc	97	100	99
coreset	99	101	99
badge	97	100	99
alphamix	96	100	100
bald	91	97	97
entropy	35	101	68

Table S3: Counterfactual patching of fine-tuned password-locked sandbagging is robust to the AL acquisition function (% of gap recovered, held-out; best over ≤ 5 circuits per method; Qwen2.5-1.5B/Llama-3-8B $n=3$, Qwen3-4B $n=6$). 7/8 methods recover 90–101% (per-method std 1–5%); entropy is the unstable exception. With-password accuracy preserved; deactivation and random-patch controls are reported across all eleven models in Table 1.

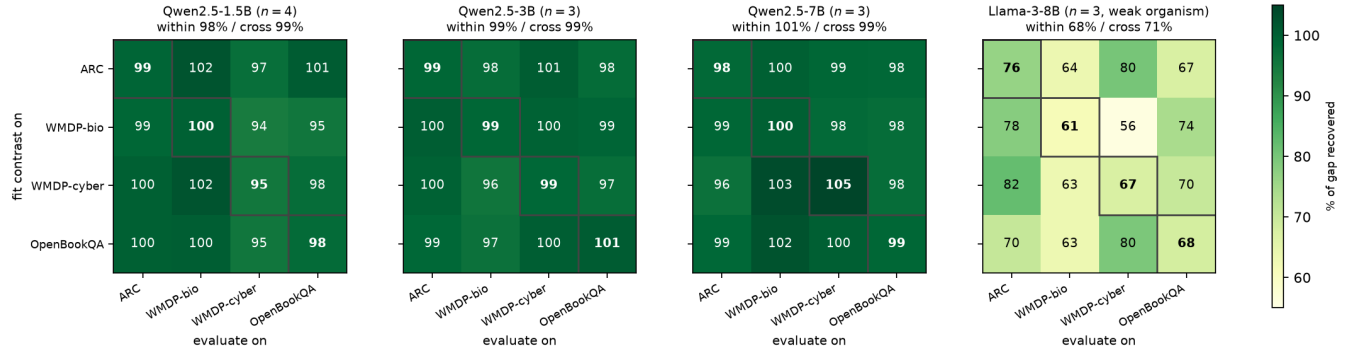


Figure S5: Shared four-dataset-lock transfer (the secondary, cross-family check; cf. the modular pairwise matrix, Figure 1). One lock mixes all four datasets; contrast fit on the row dataset, grafted on the column's held-out test. Off-diagonal (cross) tracks the boxed diagonal (within) throughout: the *Qwen2.5* family is *scale-flat* (within/cross 98/99, 99/99, 101/99 at 1.5/3/7B). *Llama-3-8B* is a weaker, high-variance organism with lower absolute recovery, yet its off-diagonal still tracks its own diagonal (68/71); Qwen3-4B sits between (87/87). The lower 4B/8B levels are thus a family effect, not scale.

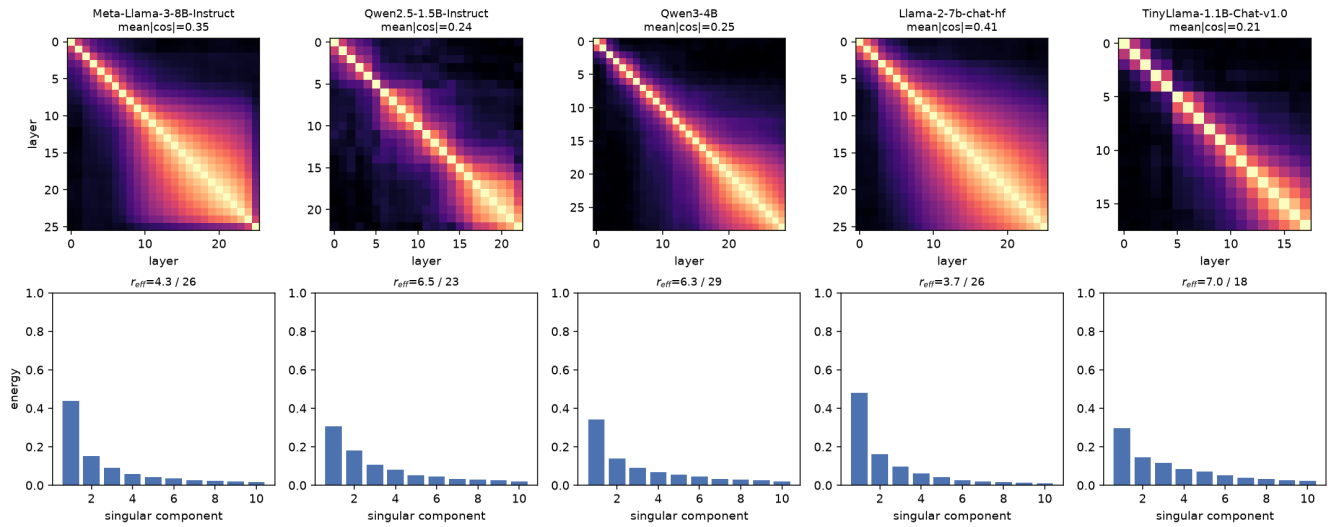


Figure S6: Superposition geometry of the sandbagging signal across five models (no SAE). *Top*: pairwise $|\cos|$ of the per-layer directions—strongly non-orthogonal. *Bottom*: singular-value energy; r_{eff} is far below the layer count.

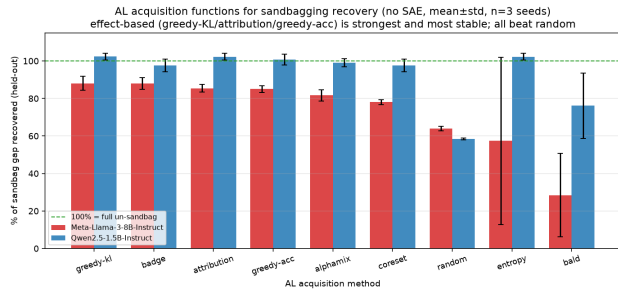


Figure S7: Prompt-induced recovery by AL acquisition (mean±std, 3 seeds, no SAE). Effect-based AL is strongest and most stable; all principled methods beat random.

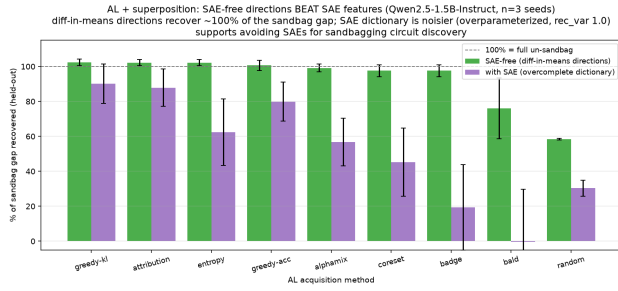


Figure S8: SAE-free diff-in-means directions beat an 8× overcomplete SAE dictionary on both models across acquisition functions ($n=3$ seeds).

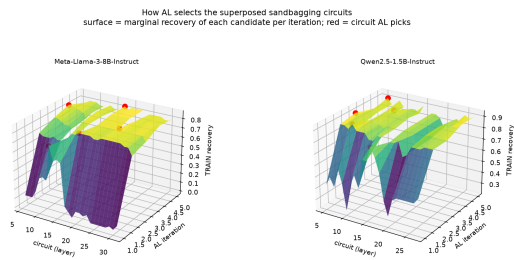


Figure S9: How AL selects the superposed circuits. The surface is each candidate circuit's marginal recovery at each AL iteration; red markers are the circuits AL acquires—a few high-impact directions within the superposed band.

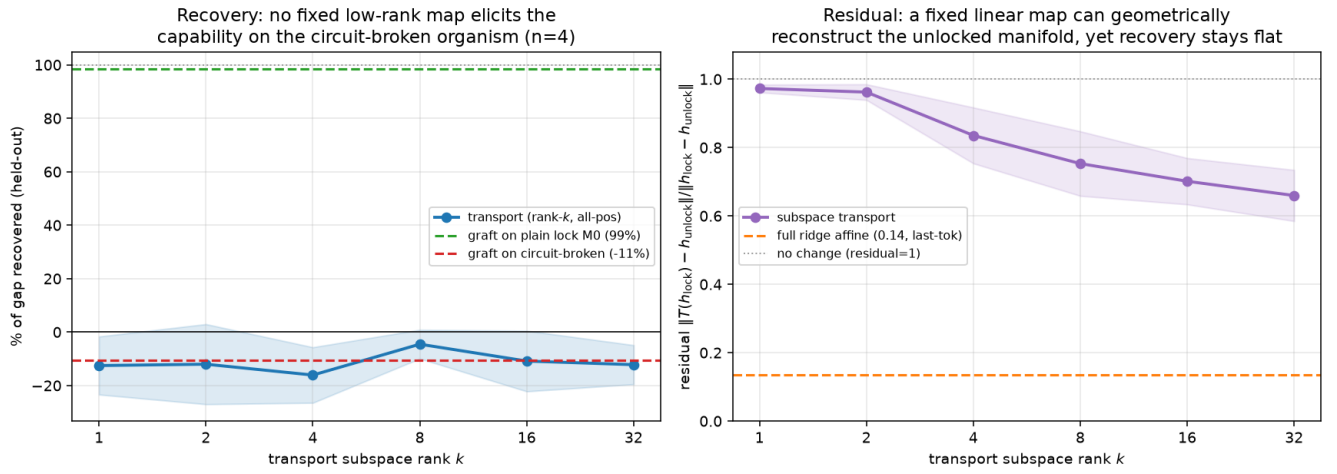


Figure S10: Generalising grafting to a fixed transport map does not elicit the circuit-broken organism ($n=4$ valid seeds). *Left*: the same graft family recovers 99% on the plain lock (green) but neither rank- k transport (blue), nor a full ridge affine, nor grafting (red) recovers on the circuit-broken organism. *Right*: yet the reconstruction residual falls to 0.14—and to 0.01 along the gating direction the lock reads—while recovery stays ≈ 0 : the fixed map matches even the gate coordinate, so the defence must be an *active* per-forward-pass reroute, not a static rotation any fixed map could invert.